

PHYSICS 1

UNIT 2: FORCES, NEWTON'S LAWS, STATICS & ENERGY

Fall 2026

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Name: _____

How to Use This Packet

Each section below has two parts, in the Cornell note-taking format, plus a summary box.

Notes column (right, wide)

Fill this in during lecture. Definitions, formulas, and key terms are partly blanked — write the missing words, symbols, or numbers as your instructor presents them.

Cue column (left, narrow)

These are the key questions and terms for the section. Use them right after class (or before a quiz) to test yourself: cover the notes column and see if you can answer each cue from memory.

Summary box

At the end of each section, write 2–3 sentences in your own words capturing the main idea. If you can't summarize it, you don't know it yet — go back and review.

Chapter 4: Dynamics — Force and Newton's Laws of Motion

4.1 Development of Force Concept

CUE / KEY QUESTIONS <i>What is dynamics?</i> <i>Definition of force — what kind of quantity is it?</i> <i>What is a free-body diagram?</i>	Dynamics is the study of how _____ affect the motion of objects. A force is a _____ or _____ that has both magnitude and direction — meaning it is a _____ quantity. Forces that act on a body from outside it are called _____ forces. A _____ is a drawing showing every external force acting on a single body, represented as arrows. Sketch a free-body diagram here for a book resting on a table:
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

4.2 Newton's First Law of Motion: Inertia

CUE / KEY QUESTIONS <i>State Newton's First Law, word for word.</i> <i>What is inertia?</i> <i>Define mass.</i>	Newton's First Law (the “Law of _____”): a body at rest remains at rest, and a body in motion remains in motion at a constant _____, unless acted on by a net _____ force. Inertia is the tendency of an object to _____. Mass is the quantity of _____ in an object, and it is what determines an object's _____ — its resistance to a change in motion.
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

4.3 Newton's Second Law of Motion: Concept of a System

CUE / KEY QUESTIONS <i>State Newton's Second Law, in words and as an equation.</i> <i>Define acceleration.</i> <i>What is weight? Give the equation.</i> <i>What is free fall?</i> <i>Define friction.</i>	Acceleration is a change in _____ — this can be a change in _____, _____, or both. Newton's Second Law: the acceleration of a system is directly proportional to the net _____ force acting on it, and inversely proportional to its _____. In equation form: $a = \frac{F_{\text{net}}}{m}$ rearranged: $F_{\text{net}} = ma$ Weight is the force of _____ on a mass: $w = mg$. If the only force acting on an object is gravity, the object is said to be in _____. Friction is a force that opposes the _____ of objects that are touching.
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

4.4 Newton's Third Law of Motion: Symmetry in Forces

CUE / KEY QUESTIONS <i>State Newton's Third Law.</i> <i>Give your own example of an action–reaction pair.</i> <i>What is thrust?</i>	Newton's Third Law: whenever one body exerts a force on a second body, the second body exerts a force back on the first that is equal in _____ and opposite in _____. My own example of an action–reaction pair: _____ A _____ is a reaction force that pushes a body forward in response to a backward-directed force (e.g., rocket exhaust).
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

4.5 Normal, Tension, and Other Examples of Forces

CUE / KEY QUESTIONS

*Define normal force —
which direction does it act?*

Define tension.

*Weight components on a
frictionless incline at angle
 θ ?*

The _____ force (N) is the support force exerted by a surface, always directed _____ to that surface.

On a frictionless incline at angle θ : $N =$ _____ $w_{\parallel}$ (along incline) = _____ $w_{\perp}$ (into incline) = _____

Tension (T) is the pulling force along a stretched flexible connector, such as a rope or cable. For a mass hanging at rest: $T =$ _____

SUMMARY — in your own words, summarize this section in 2–3 sentences:

Chapter 5: Further Applications of Newton's Laws — Friction, Drag, and Elasticity

5.1 Friction

CUE / KEY QUESTIONS <i>Define friction.</i> <i>Static vs. kinetic friction — what's the difference?</i> <i>Static friction inequality?</i> <i>Kinetic friction equation?</i>	Friction is a _____ force that opposes _____ or attempted motion between two surfaces in contact. _____ friction acts between surfaces that are NOT moving relative to each other; _____ friction acts between surfaces that ARE sliding relative to each other. Static friction: $f_s \leq$ _____ N, where μ_s is the coefficient of _____ friction. Kinetic friction: $f_k =$ _____, where μ_k is the coefficient of _____ friction.
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

5.2 Drag Forces

CUE / KEY QUESTIONS <i>What is drag?</i> <i>Drag force equation (large objects)?</i> <i>Stokes' Law — when does it apply, and what's the equation?</i>	_____ forces oppose the motion of an object moving through a fluid (air or liquid). For larger, faster-moving objects: $F_D =$ _____, where C is the _____, ρ is fluid density, A is cross-sectional area, and v is speed. Stokes' Law (small objects in a viscous fluid): $F_s =$ _____, where r is the object's radius and η is the fluid's _____.
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

5.3 Elasticity: Stress and Strain

CUE / KEY QUESTIONS <i>State Hooke's Law.</i> <i>Define stress and strain.</i> <i>Name the 3 elastic moduli and what each governs.</i>	Hooke's Law: $F = \underline{\hspace{2cm}}$ (valid as long as the deformation is not too large). $\underline{\hspace{2cm}}$ is the ratio of force to area (F/A); $\underline{\hspace{2cm}}$ is the ratio of change in length to original length ($\Delta L/L_0$), and is unitless. The three elastic moduli: $\underline{\hspace{2cm}}$ modulus (Y) — stretch/compression $\underline{\hspace{2cm}}$ modulus (S) — shear/sideways deformation $\underline{\hspace{2cm}}$ modulus (B) — volume change.
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

Chapter 9: Statics and Torque

9.1 The First Condition for Equilibrium

CUE / KEY QUESTIONS <i>Define statics.</i> <i>State the first condition for equilibrium.</i>	_____ is the study of forces and torques in _____. First condition for equilibrium: the net _____ force on a system must equal _____. In equation form: $\Sigma F = \underline{\hspace{2cm}}$
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SUMMARY — in your own words, summarize this section in 2–3 sentences: _____ _____ _____

9.2 The Second Condition for Equilibrium

CUE / KEY QUESTIONS <i>Define torque.</i> <i>What is a lever arm?</i> <i>Sign convention for torque?</i> <i>State the second condition for equilibrium.</i>	_____ is the rotational equivalent of force — it's what causes a change in _____ motion. $\tau = rF \sin \theta$, where $r_{\perp} = r \sin \theta$ is called the _____. SI unit of torque: _____ By convention, _____ torques are positive and _____ torques are negative. Second condition for equilibrium: the net external _____ must equal zero: $\Sigma \tau = \underline{\hspace{2cm}}$
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SUMMARY — in your own words, summarize this section in 2–3 sentences: _____ _____ _____

9.3 Stability

CUE / KEY QUESTIONS <i>Define the 3 types of equilibrium: stable, unstable, neutral.</i>	_____ equilibrium: a small displacement creates a restoring force/torque that pushes the object back toward equilibrium. _____ equilibrium: a small displacement creates a force/torque that increases the displacement even further. _____ equilibrium: a displacement creates no net force or torque at all — the object simply stays in its new position.
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

9.4 Applications of Statics

CUE / KEY QUESTIONS <i>What makes statics a “special case” of Newton's Laws?</i> <i>What are the recommended problem-solving steps?</i>	Statics is a special case of Newton's Laws where _____ = 0 and _____ = 0 at the same time — an object is at rest, or moving at constant velocity, with no rotation. Problem-solving tip: sketch a _____ first, choose a convenient _____ point, and apply both conditions for equilibrium.
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

Chapter 7: Work, Energy, and Energy Resources

7.1 Work: The Scientific Definition

CUE / KEY QUESTIONS <i>Define work (the physics definition).</i> <i>Equation for work?</i> <i>SI unit of work?</i>	Work is the transfer of _____ by a force acting on an object as it is _____. $W = \text{_____}$, where θ is the angle between the force and the displacement. If the force and displacement point the same direction, $\theta = \text{_____}^\circ$ and W is at its _____. If the force is perpendicular to the displacement, $W = \text{_____}$. SI unit of work: the _____, where $1 \text{ J} = 1 \text{ N}\cdot\text{m} = 1 \text{ _____}$
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

7.2 Kinetic Energy and the Work-Energy Theorem

CUE / KEY QUESTIONS <i>Equation for kinetic energy?</i> <i>State the work-energy theorem.</i>	Kinetic energy: $KE = \text{_____}$ Work-Energy Theorem: the net work done on an object equals its change in kinetic energy: $W_{\text{net}} = \text{_____} - \text{_____}$
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

7.3 Gravitational Potential Energy

CUE / KEY QUESTIONS <i>Equation for gravitational PE?</i> <i>Relationship between ΔPE and ΔKE on a frictionless descent?</i>	Gravitational potential energy: $\Delta PE_g = \text{_____}$, where h is the height. On a frictionless descent, the loss in PE equals the gain in KE: $\Delta KE = \text{_____}$ ΔPE_g .
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

7.4 Conservative Forces and Potential Energy

CUE / KEY QUESTIONS <i>Define a conservative force. Give 2 examples.</i> <i>Equation for spring PE?</i> <i>Conservation of mechanical energy equation?</i>	A _____ force is one for which the work done depends only on the starting and ending points, not on the path taken. Examples: _____ Spring potential energy: $PE_s = \text{_____}$ Conservation of mechanical energy (no nonconservative forces present): $KE_i + PE_i = \text{_____} + \text{_____}$
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

7.5 Nonconservative Forces

CUE / KEY QUESTIONS <i>Define a nonconservative force. Give an example.</i> <i>Energy accounting equation with nonconservative work?</i>	A _____ force is one for which the work done DOES depend on the path taken. _____ is the classic example — it converts mechanical energy into _____ energy. Energy accounting with nonconservative work: $KE_i + PE_i + W_{nc} =$ _____ + _____
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

7.6 Conservation of Energy

CUE / KEY QUESTIONS <i>State the Law of Conservation of Energy.</i> <i>Equation for efficiency?</i>	The total energy of an isolated system is _____ — it can change _____, but it cannot be _____ or _____. Efficiency of a machine or process: $Eff =$ _____ / _____
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

7.7 Power

CUE / KEY QUESTIONS	Power is the _____ at which work is done, or energy is transferred or converted.
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<i>Define power.</i> <i>Equation for power?</i> <i>SI unit of power?</i> <i>1 horsepower = how many watts?</i>	P = _____ / _____ SI unit: the _____, where 1 W = 1 _____ 1 horsepower (hp) = _____ W.
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SUMMARY — in your own words, summarize this section in 2–3 sentences:
